

Edward (Ted) Ivanac

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EDUCATION	University of Michigan , Ann Arbor, MI	August 2019 - May 2024
	<i>Bachelor of Science in Engineering (BSE)</i> , Robotics <i>Bachelor of Science in Engineering (BSE)</i> , Electrical Engineering • Notable Courses: Simultaneous Localization & Mapping, Robot Operating Systems, Autonomous Vehicles • Awards: University Honors, Dean's List (May 2023, December 2023, May 2024) Michigander Scholar's Award (February 2024 - May 2024)	GPA: 3.41
TEACHING EXPERIENCE	Duderstadt Center Fellowship , Ann Arbor, MI	September 2020 - May 2024
	<i>Lead Fabrication Fellow</i> • Instituted an interdisciplinary fellowship program and training material for users in both multimedia and fabrication spaces • Developed and facilitated 4 workshops multiple times over three years on topics including Arduino, Raspberry Pi, Soldering, and CAD Design • Authored instructional guides for students on topics including Additive Manufacturing, CAD Design, and PCB Design	
	ENGR 110 , Ann Arbor, MI	August 2023 - December 2024
	<i>Instructional Aide</i> • Facilitated weekly discussion sections for ENGR 110 class, fostering an engaging learning environment for students in the Electrical and Robotics Engineering departments • Contributed to the refinement of existing course material through weekly feedback sessions on experiences with students • Effectively communicated the values and distinctive features of the Robotics department to peers, fostering a sense of community and enthusiasm for the field	
	Michigan Engineering Zone , Detroit, MI	September 2020 - May 2022
	<i>Lead Robotics Mentor Coordinator</i> • Created an educational robotics kit and curriculum that taught C programming as well as using encoders in a control loop through Arduino robotics hardware • Led a 20-person mentor group to create technical and non-technical workshops for high school students in the FIRST Robotics Competition	
PROFESSIONAL EXPERIENCE	General Motors , Warren, MI	July 2024 - Present
	<i>Assistant Vehicle Systems Engineer</i> • Develops Corporate Common Components plans that ensure maximum use across vehicle programs and architectures. Ensure each component is integrated and at the right cost/risk profile • Drives completion of associated business case impacts by working with interdisciplinary teams of specialists in tech, manufacturing, materials, and design along with potential buyers and PPCO	
	Ford Motor Company , Dearborn, MI	May 2022 - July 2022
	<i>ADAS Product Development Intern</i> • Developed four ADAS data analysis tools in MATLAB to post-process vehicle features success metrics specifically for the Lane Change Assist feature, surpassing target goal by three • Implemented and verified pre-made rapid-install device with a built-in GPS and IMU to track vehicle dynamics data on autonomous vehicles, resulting in more efficient methods of instrumenting vehicles	

PROJECTS

Brushless DC Motor Driver

Robot Sensors & Signals Course

April 2024

- Codeveloped brushless motor driver for use in the MBot platform with a team of three other robotics engineering students
- Redesigned open-source SimpleFOC Arduino shield as a PCB to be compatible with Raspberry Pi Pico microcontroller
- Designed test fixtures in Fusion 360 for PCB board and gimbal brushless motor to test torque control of the motor using an absolute magnetic encoder

Forward/Inverse Kinematics

Robot Operating Systems Course

January 2024 - April 2024

- Implemented forward kinematics in Javascript to map joint/link configurations to end effector poses URDF-style Javascript robot simulations
- Implemented inverse kinematics in Javascript to infer joint states to reach a desired end effector pose
- Developed Finite State Machine to cycle robot description through a series of end effector poses to create a robot dance routine

MBot SLAM & Navigation

SLAM & Navigation Course

September 2023 - December 2023

- Developed a program in C++ that generates a map in ROS2 from live LIDAR data
- Integrated a particle filter algorithm for real-time localization inside of an environment for simultaneous localization and mapping
- Implemented A* path planning algorithms to navigate a mobile robot through a maze autonomously

Autonomous Vehicle Simulation

Autonomous Vehicles Course

September 2023 - December 2023

- Implemented a Kinematic Bicycle model in MATLAB with realistic force modeling including aerodynamic drag and tire rolling resistance
- Developed and tuned PI controllers for velocity control and steering wheel control to autonomously navigate the vehicle along a series of predetermined waypoints
- Implemented A* path planner library into a simulation model to plan a path and navigate through a map with randomly generated obstacles

CERTIFICATION

The Arduino Platform and C Programming by University of California, Irvine on Coursera *Intro course to the Arduino platform and using the C programming language*
Verify : <https://coursera.org/verify/N439REZ9PFVB>

TECHNICAL SKILLS

Software : C++, MATLAB, Python, Simulink, Javascript, Github, Stateflow, Julia, Fusion 360, EAGLE CAD

Hardware : PCB Design, 3D Printing, Raspberry Pi Pico, Lidar, CAN, IMUs